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Forecasting Danish Electricity Prices

Machine Learning Approach Using Market Fundamentals

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Abstract

Electricity prices in Denmark are highly volatile due to the interaction of demand fluctuations, renewable generation, fuel costs and cross-border electricity flows. Understanding these dynamics is essential for electricity traders, utilities and energy market analysts operating in the Nordic and European power markets.

This project develops a complete data pipeline to analyse and forecast Danish day-ahead electricity prices in the two bidding zones DK1 (West Denmark) and DK2 (East Denmark). The workflow covers data ingestion from the Danish Energy Data Service API, exploratory analysis, feature engineering and the implementation of machine learning forecasting models. Three modelling approaches are evaluated: a baseline time-series model based on historical prices, a model incorporating key market fundamentals and a broader structural model including additional electricity system variables.

The results show that incorporating market fundamentals significantly improves forecasting accuracy compared with a pure time-series approach. In particular, wind generation, temperature and gas prices provide valuable predictive signals because they reflect the main drivers of supply, demand and marginal generation costs in the Danish electricity market. While additional structural variables such as load and cross-border flows increase the economic richness of the model, they do not further improve forecasting performance.

Overall, the analysis demonstrates that a focused model combining time-series dynamics with a small set of economically meaningful market drivers provides the best balance between predictive accuracy and interpretability. The project highlights how data analytics and machine learning techniques can be applied to energy markets to support electricity price analysis and short-term forecasting.

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1. Introduction

Electricity markets are characterised by high volatility because supply and demand must remain balanced at every moment. In Denmark, price dynamics are strongly influenced by renewable generation, cross-border electricity trading and weather-driven demand fluctuations.

Forecasting electricity prices is therefore an important task for energy market participants. Accurate forecasts help market analysts evaluate price dynamics, support electricity trading strategies and improve market monitoring.

This project develops a structured forecasting pipeline to analyse Danish day-ahead electricity prices in DK1 and DK2. The modelling process evolves through three stages in order to evaluate how additional economic information improves forecasting performance.

1.1 Data and Pipeline Architecture

The dataset used in this project is obtained from the Danish Energy Data Service API. Electricity prices are retrieved from the Elspotprices dataset and combined with more recent observations from the DayAheadPrices dataset.

The ingestion process is implemented using Python scripts that automatically download the data and store it locally using DuckDB. This provides a lightweight analytical database suitable for local data pipelines.

The pipeline consists of several stages:

- Data ingestion from the API
- Storage and querying using DuckDB
- Exploratory data analysis
- Feature engineering
- Model training and evaluation
- Forecast generation

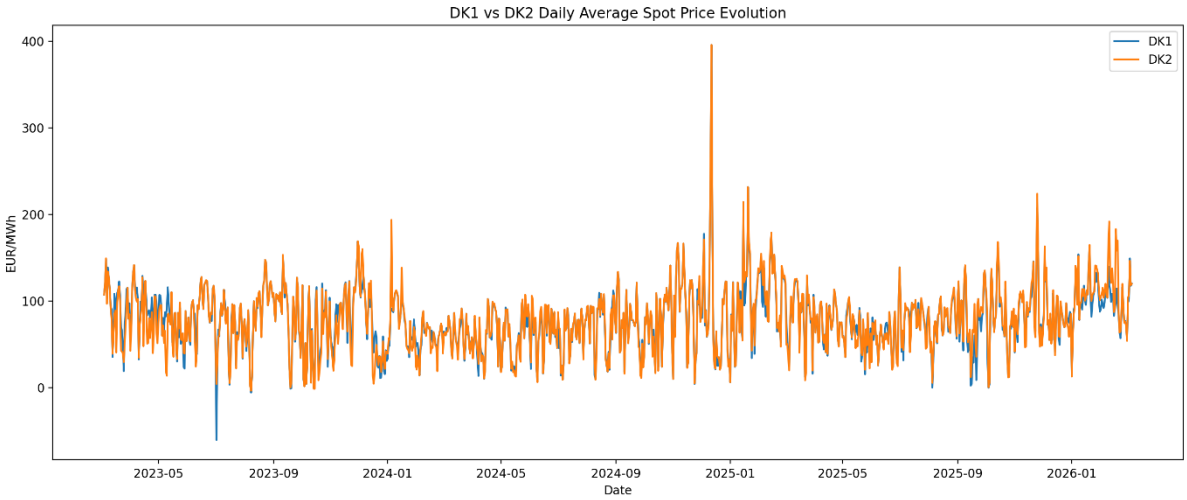
This modular pipeline structure ensures that the project can be reproduced and extended easily.

1.2 Exploratory Analysis

Exploratory analysis reveals that Danish electricity prices display strong volatility and frequent price spikes. These characteristics are typical of electricity markets where small changes in supply or demand can produce large price movements.

DK1 and DK2 generally follow similar long-term trends due to the integration of the Danish electricity system within the Nordic and European power markets. However, persistent differences between the two zones indicate that local conditions and transmission constraints still influence price formation.

Figure 1. Historical electricity price evolution in DK1 and DK2



2. Baseline Time-Series Model

The first modelling stage introduces a baseline forecasting model based exclusively on time-series information.

The model uses calendar variables including hour of day, day of week and month in order to capture recurring consumption patterns. Lagged price variables are also included to represent short-term persistence and weekly repetition in electricity prices.

Rolling statistics such as moving averages and standard deviations capture recent market volatility and trends.

This baseline model therefore relies entirely on historical price behaviour. Although it does not incorporate economic fundamentals, it provides a useful benchmark for evaluating more advanced models.

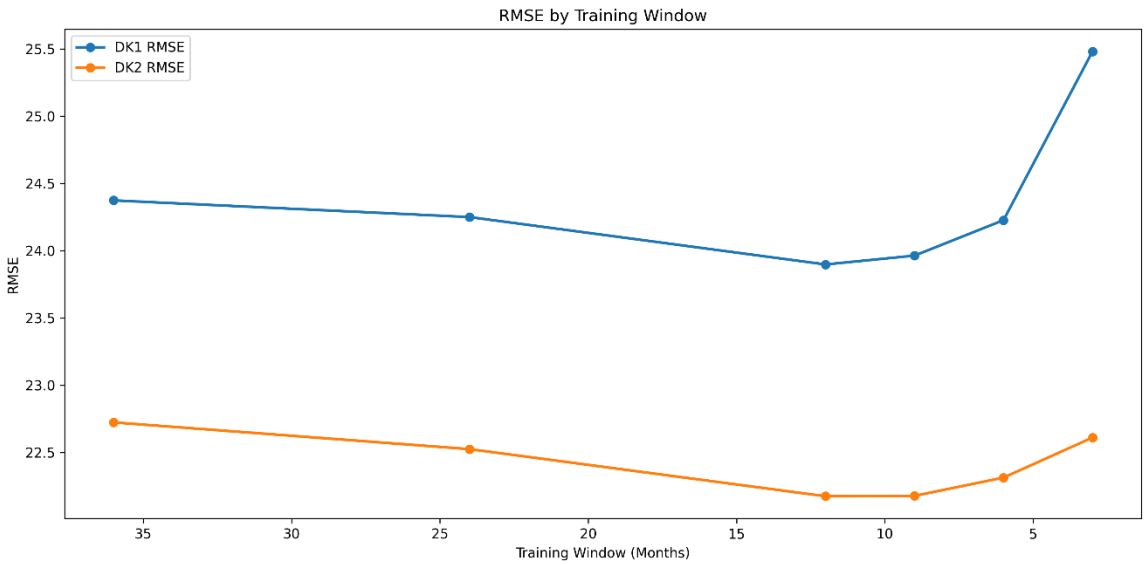
2.1 Training Window Optimisation

A key modelling decision concerns the amount of historical data used to train the model.

Using too little history can lead to unstable forecasts, while using very long historical windows may reduce performance because electricity market conditions evolve over time.

Several training windows were therefore evaluated using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE).

Figure 2. Model performance across training window



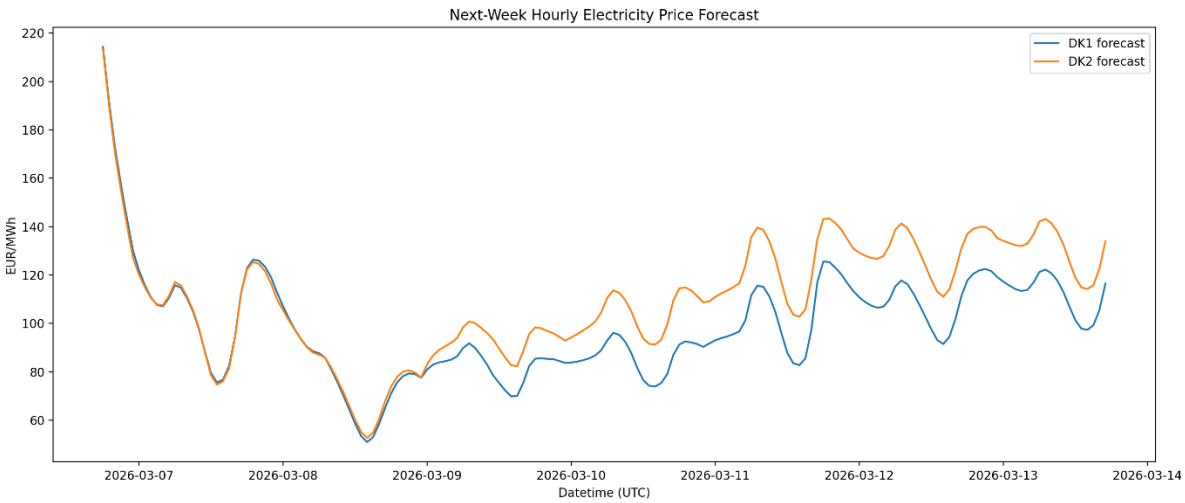
The results show that a **12-month training window** provides the best performance for the baseline model. This window captures a complete seasonal cycle while remaining representative of recent market conditions.

2.2 Forecast Generation

The final baseline model is used to generate hourly electricity price forecasts for the next week.

These forecasts capture general temporal patterns in electricity prices, although they do not yet include explicit economic drivers such as renewable production or fuel costs.

Figure 3. Next week electricity price forecast



3. Market Drivers Model

The second modelling stage incorporates key market fundamentals that influence electricity price formation.

Three variables are introduced:

Wind generation, Temperature, Gas prices

Wind generation represents renewable electricity supply and plays a particularly important role in Denmark due to the country's high wind penetration.

Temperature is used as a proxy for electricity demand, as heating and cooling needs vary with weather conditions.

Gas prices represent marginal generation costs in thermal power plants and therefore influence electricity prices through production costs.

3.1 Model Performance

The introduction of market fundamentals significantly improves forecasting performance in both DK1 and DK2.

Table 1. Forecasting performance across models

Model	DK1 MAE	DK1 RMSE	DK2 MAE	DK2 RMSE
Baseline Model	10.24	18.14	11.49	21.19
Market Drivers Model	*7.55*	*13.36*	*8.68*	*15.79*

These results demonstrate that electricity prices respond strongly to supply conditions, weather-driven demand and fuel costs.

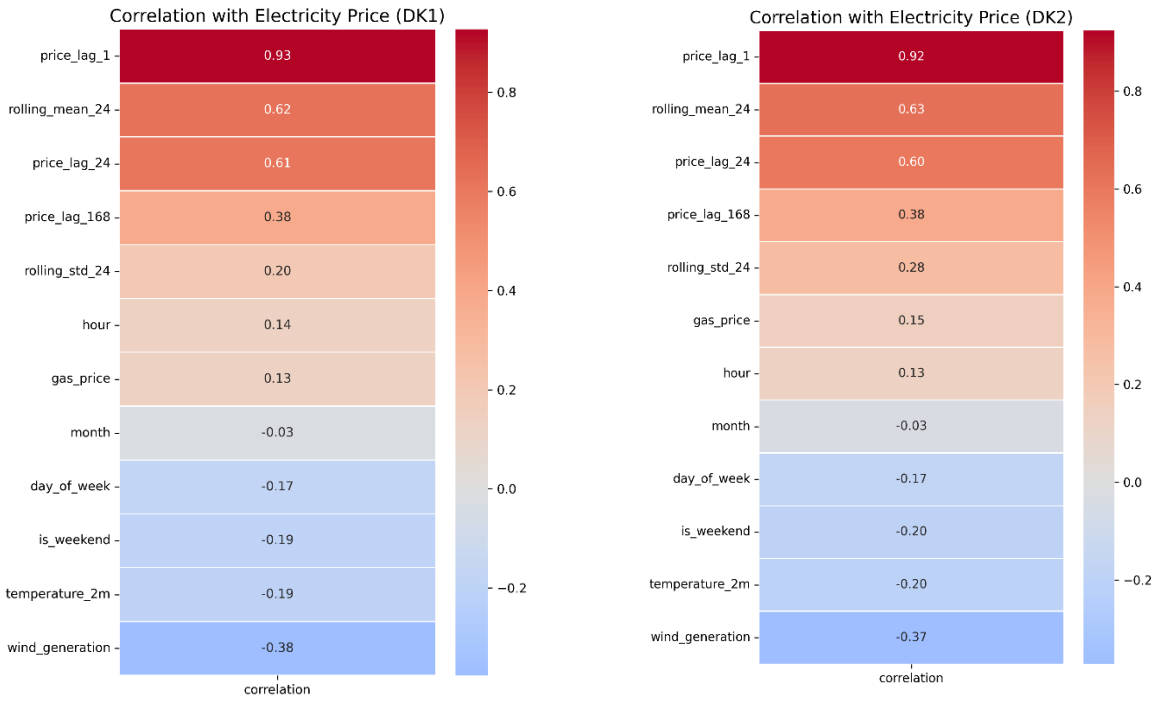
From an economic perspective, incorporating market fundamentals improves both predictive accuracy and model interpretability.

3.2 Feature Relationships with Price

To better understand the economic drivers behind the forecasting model, correlation analysis was conducted between electricity prices and the explanatory variables included in the Market Drivers model.

Correlation analysis provides insight into how changes in key market variables are associated with movements in electricity prices. In electricity markets, these relationships are often driven by supply dynamics, demand conditions and marginal generation costs.

Figure 4. Correlation between variables and electricity price (DK1 & DK2)



The correlation results are economically intuitive. Wind generation tends to show a negative relationship with electricity prices because higher renewable output increases electricity supply and reduces the need for more expensive generation sources. Temperature also influences electricity demand, particularly during colder periods when heating demand increases.

Gas prices are expected to show a positive relationship with electricity prices, since gas-fired power plants often determine the marginal cost of electricity generation in European power markets. When gas prices increase, the cost of producing electricity rises, which can push electricity prices upward.

These relationships highlight the importance of incorporating market fundamentals into electricity price forecasting models.

3.3 Actual vs Predicted Prices

To evaluate the performance of the forecasting model, predicted electricity prices are compared with the observed market prices. This comparison allows us to visually assess how closely the model reproduces real price dynamics in the Danish electricity market.

Figure 6. Actual vs predicted electricity prices (DK1)

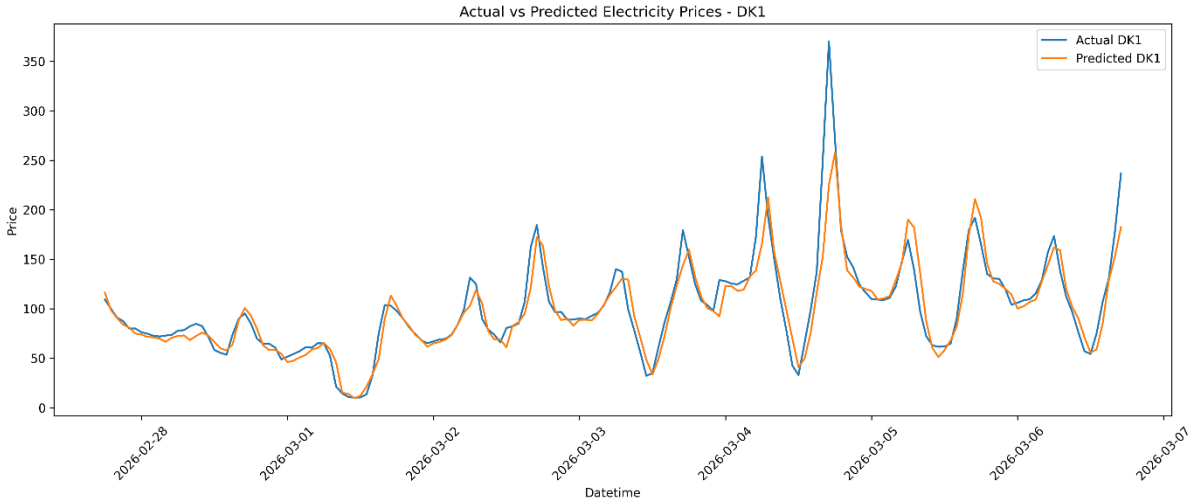
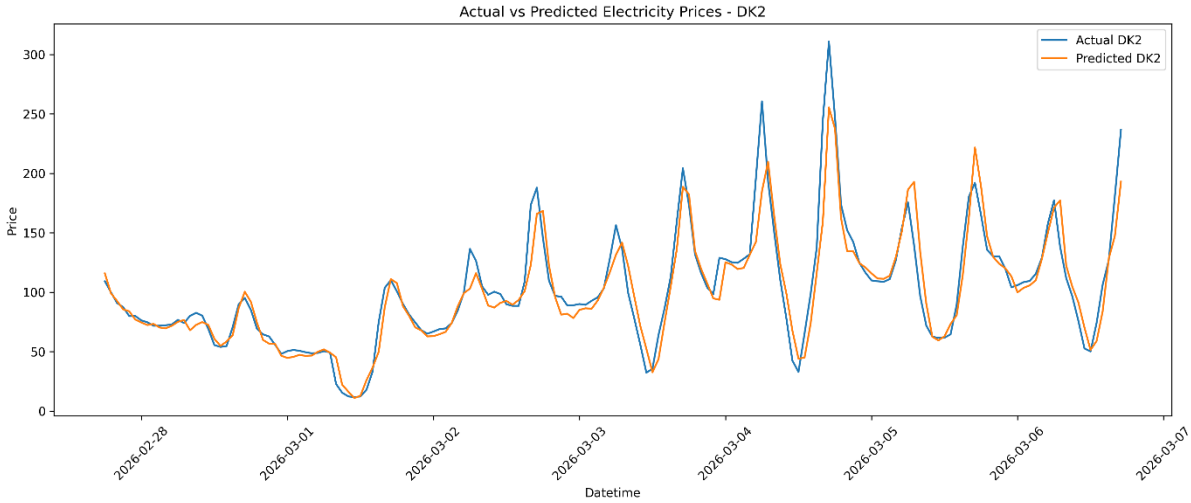


Figure 7. Actual vs predicted electricity prices (DK2)



The comparison between observed and predicted prices shows that the Market Drivers model captures the overall dynamics of electricity prices relatively well. The model successfully reproduces the general level of prices and many short-term fluctuations.

However, some extreme price spikes are more difficult to predict accurately. This behaviour is common in electricity price forecasting because sudden supply or demand shocks, unexpected renewable generation changes or transmission constraints can generate large price movements that are difficult to anticipate using statistical models.

Despite these limitations, the model provides a reasonable approximation of market behaviour and demonstrates that incorporating market fundamentals significantly improves the ability to forecast electricity prices.

4. Extended Structural Model

The final stage introduces additional structural variables including electricity load, cross-border electricity flows and price signals from neighbouring markets.

These variables attempt to capture broader electricity market dynamics, particularly the influence of regional electricity trading and physical grid constraints.

However, the results indicate that including these additional variables does not further improve forecasting accuracy. Although the model remains strong, the additional complexity introduces noise and reduces predictive performance relative to the Market Drivers model.

4.1 Model Performance

To evaluate the performance of the forecasting models, prediction errors are measured using **Mean Absolute Error (MAE)** and **Root Mean Squared Error (RMSE)**. These metrics quantify the difference between predicted electricity prices and observed market prices.

Table 1. Forecasting performance across models

Model	DK1 MAE	DK1 RMSE	DK2 MAE	DK2 RMSE
Baseline Model	10.24	18.14	11.49	21.19
Market Drivers Model	*7.55*	*13.36*	*8.68*	*15.79*
Improve Market Drivers	8.10	13.83	8.88	16.58

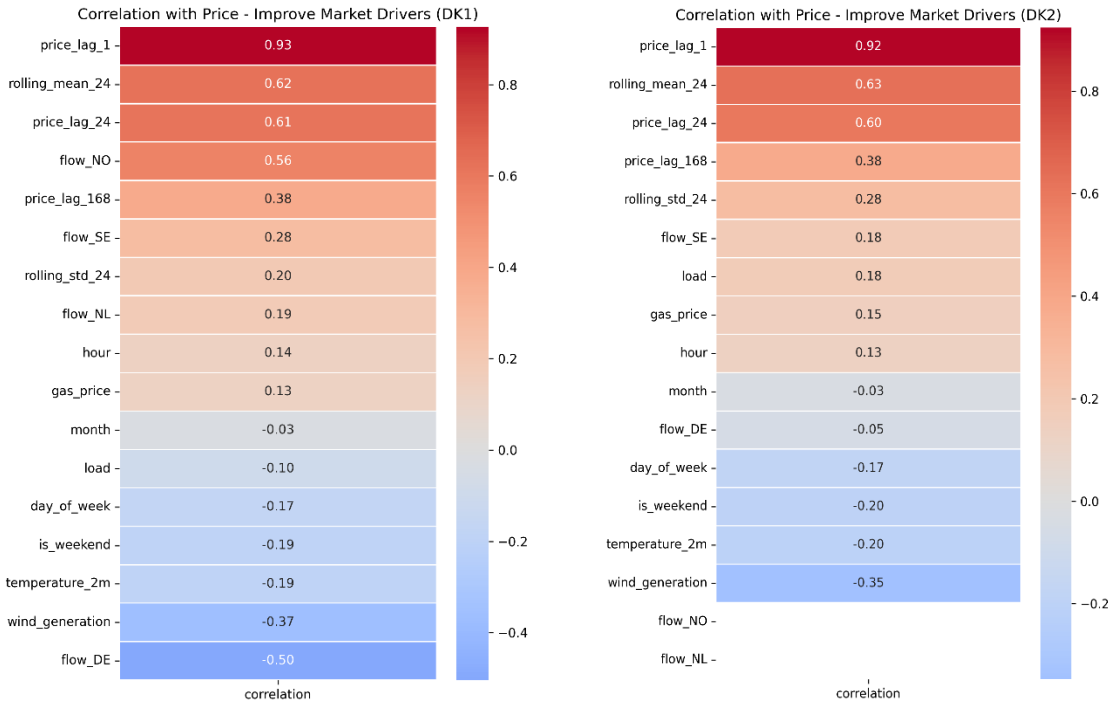
The results show that incorporating market fundamentals significantly improves forecasting accuracy compared with the baseline time-series model. In both Danish bidding zones, the Market Drivers model produces the lowest forecasting errors.

This improvement suggests that electricity prices respond strongly to real market conditions such as renewable generation, demand fluctuations and fuel costs. Although the improved structural model includes additional variables, it does not outperform the Market Drivers model, indicating that a smaller set of economically meaningful variables may provide the most efficient forecasting framework.

4.2 Feature Relationships with Price

To better understand the economic drivers of the forecasting models, correlation analysis was conducted between electricity prices and the explanatory variables included in the models.

Figure 6. Correlation between variables and electricity price (DK1 & DK2)



The correlation analysis reveals relationships that are consistent with electricity market theory. Wind generation typically shows a negative relationship with electricity prices because higher renewable output increases electricity supply and reduces the need for more expensive generation technologies.

Gas prices tend to show a positive relationship with electricity prices since gas-fired power plants often determine the marginal cost of electricity generation in European electricity markets. Temperature also influences demand conditions, which can affect electricity price dynamics.

These relationships confirm that incorporating market fundamentals improves the economic interpretability of the forecasting model.

4.3 Actual vs Predicted Prices

To visually evaluate model performance, predicted electricity prices are compared with observed prices.

Figure 8. Actual vs predicted prices (DK1)

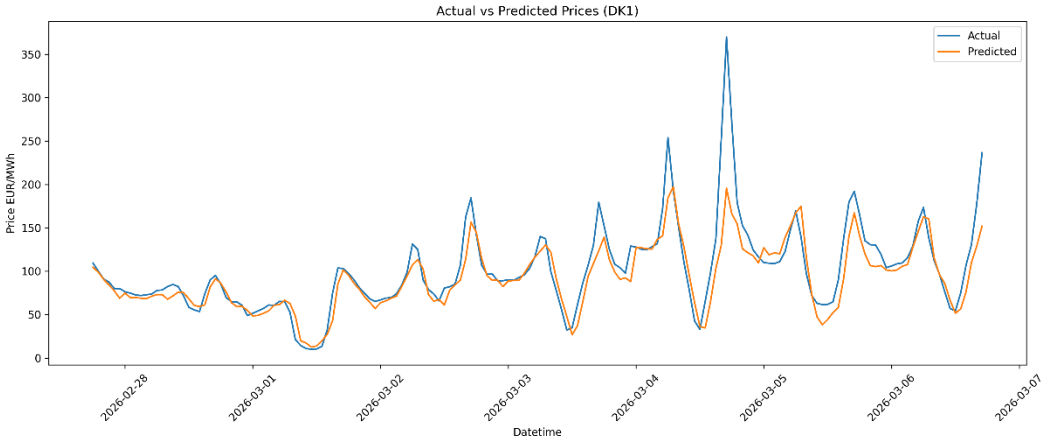
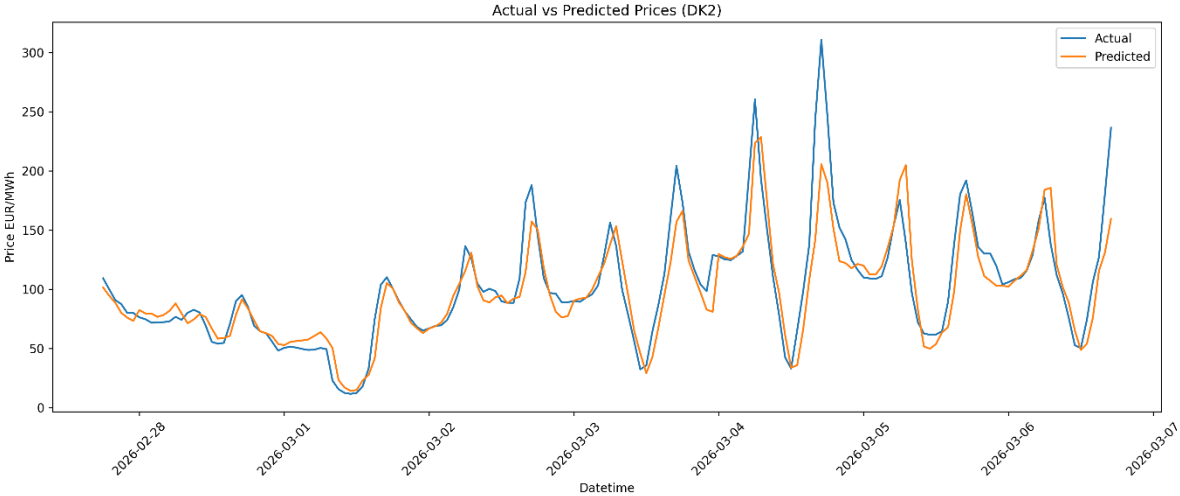


Figure 9. Actual vs predicted prices (DK2)



Although the improved model incorporates additional structural information about the electricity system, its predictive performance remains slightly lower than that of the simpler Market Drivers model. This suggests that the additional variables introduce complexity and potential noise without providing sufficient additional predictive power.

The results therefore highlight the importance of balancing model complexity with predictive efficiency when developing electricity price forecasting models.

5. Forecast Comparison Across Models

To better understand the differences between the forecasting approaches, the predictions generated by each model are compared across the Danish electricity price zones DK1 and DK2.

Figure 12. Forecast comparison across models (DK1)

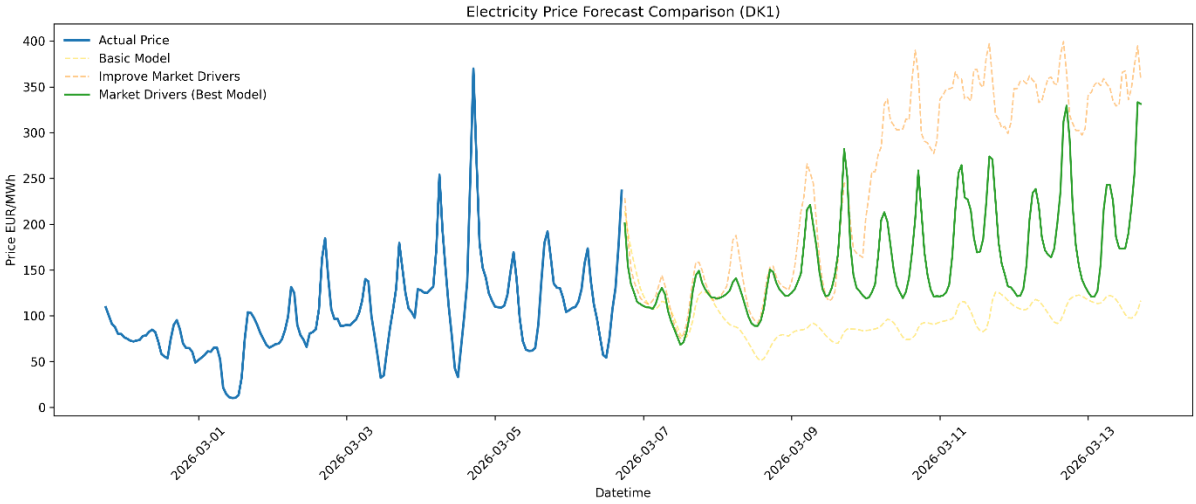
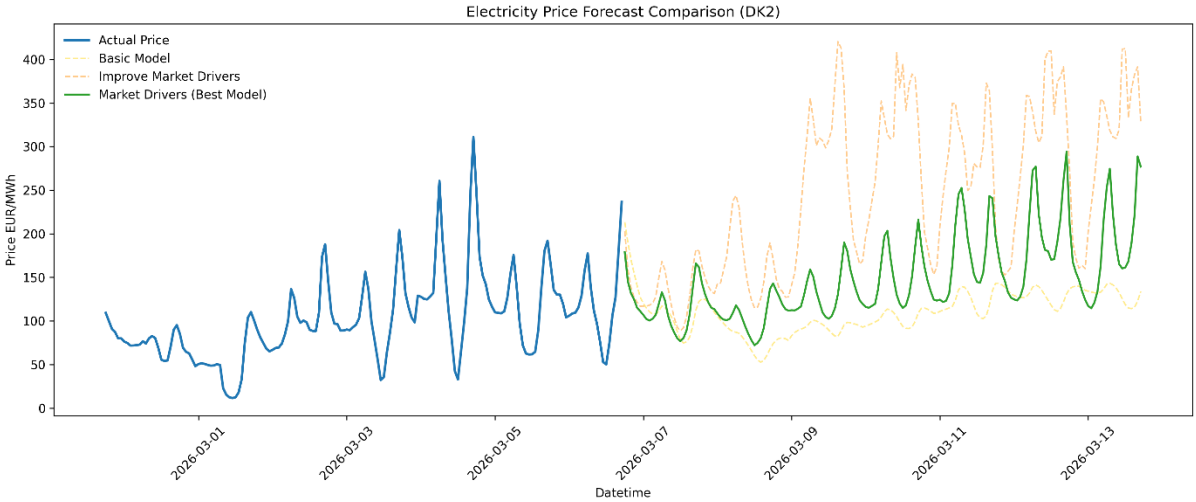


Figure 13. Forecast comparison across models (DK2)



The comparison highlights the differences in how each model captures electricity price dynamics. The baseline time-series model is able to reproduce general temporal patterns but lacks sensitivity to real market conditions.

The Market Drivers model performs more consistently, capturing both the overall price level and many short-term fluctuations. By incorporating wind generation, temperature and gas prices, the model responds more effectively to supply and demand dynamics.

The extended structural model introduces additional variables such as electricity load and cross-border flows. While these variables add economic richness, they do not significantly improve predictive performance and may introduce additional noise into the forecasting process.

Overall, the comparison suggests that the Market Drivers model offers the most reliable forecasts among the three modelling approaches.

6. Conclusion

This project developed a complete data pipeline for analysing and forecasting Danish day-ahead electricity prices in the two bidding zones DK1 and DK2. The workflow combined data ingestion from the Danish Energy Data Service API, feature engineering, machine learning models and visual analysis of forecasting performance.

Three modelling approaches were evaluated. The baseline model relied solely on historical price information and time-series features. The second model introduced key market fundamentals including wind generation, temperature and gas prices. The final model incorporated additional structural variables such as electricity load, cross-border electricity flows and neighbouring electricity markets.

The results demonstrate that incorporating market fundamentals significantly improves forecasting accuracy compared with a purely autoregressive time-series model. Wind generation, temperature and gas prices provide valuable predictive signals because they capture the fundamental drivers of electricity supply, demand and marginal generation costs.

Although the extended structural model adds additional economic variables, its predictive performance does not exceed that of the Market Drivers model. This suggests that a more parsimonious model using a focused set of economically meaningful variables may provide the best balance between predictive accuracy and interpretability.

Overall, the analysis shows that combining time-series techniques with market fundamentals provides a robust approach to short-term electricity price forecasting in the Danish power market.